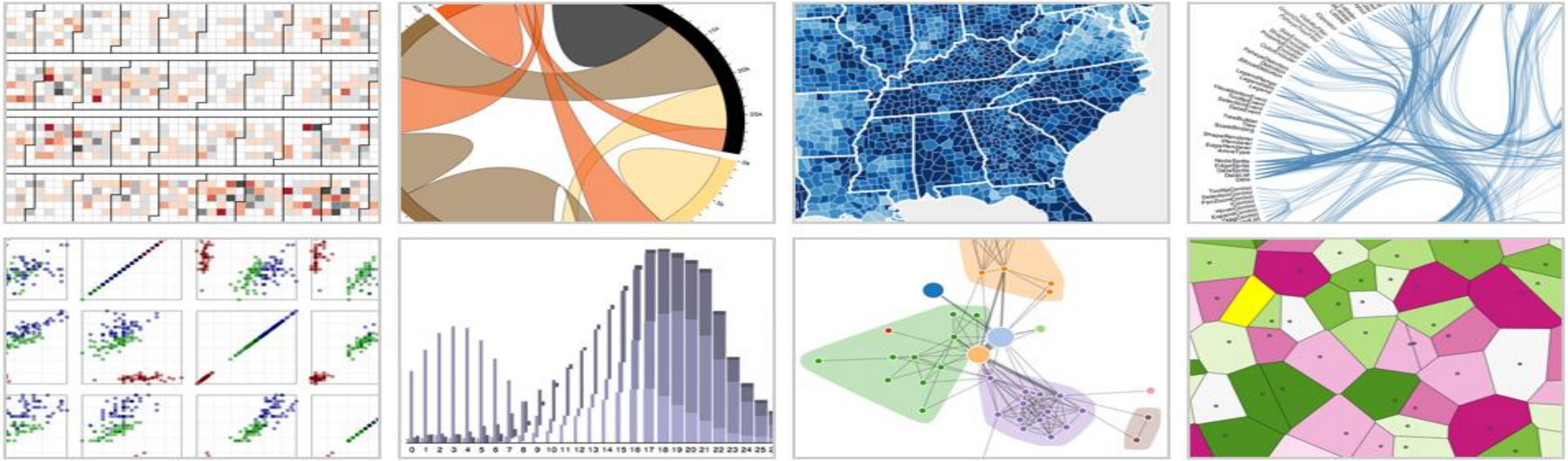
CDS6324
DATA VISUALIZATION



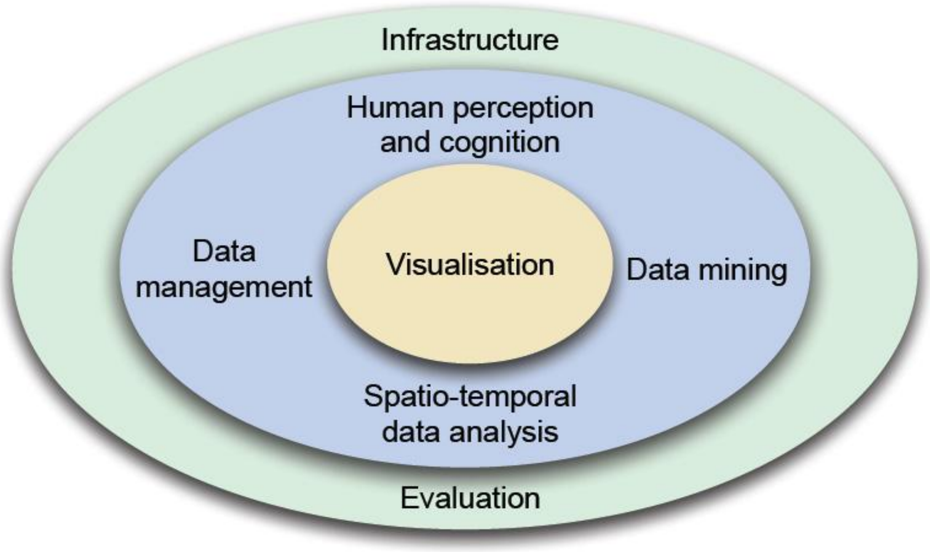
Lecture 13: Visualization Applications in Industry

CDS 6324
DATA VISUALIZATION

Lecture 12: Visualization Applications in Industry

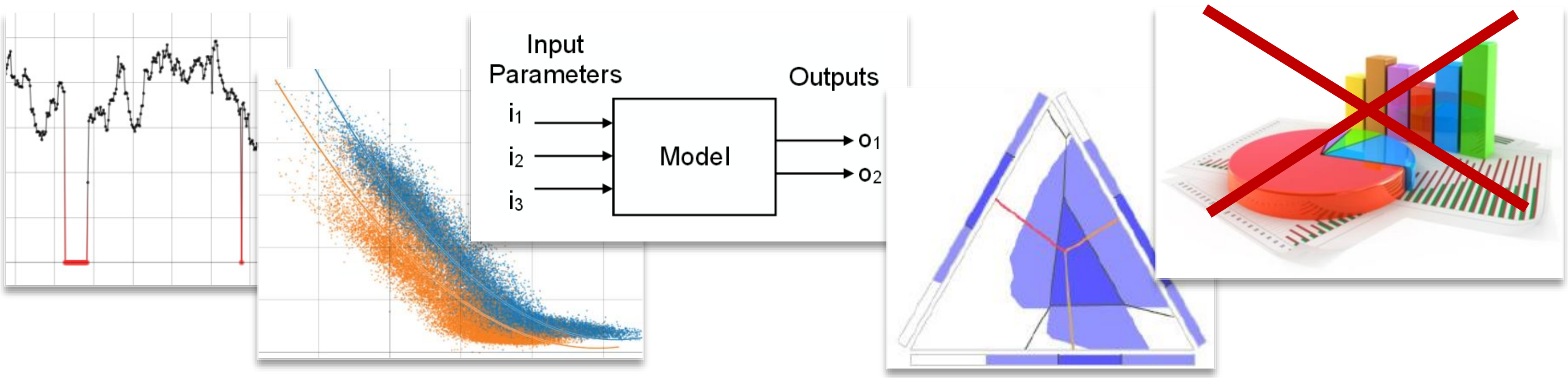
Visual Analytics

- … „ combines automated analysis techniques with interactive visualizations for an effective understanding, reasoning and decision making on the basis of very large and complex datasets"
  - [D. Keim, J. Kohlhammer, G. Ellis, F. Mansmann. "Mastering the information age: Solving problems with visual analytics". VisMaster, 2010.]



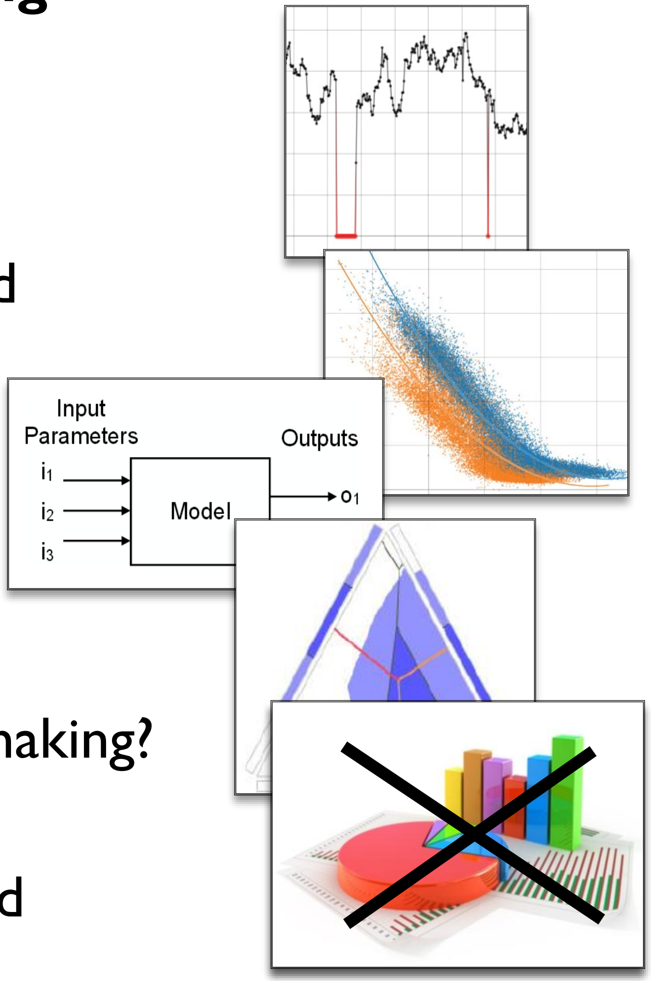
Recurring Questions …

- Is the data quality sufficient for correct interpretation?
- Does the data reveal unexpected trends and relationships?
- Does the data enable to build (predictive) models?
- How to use the data for efficient decision making?
- How to effectively communicate insights and decisions?



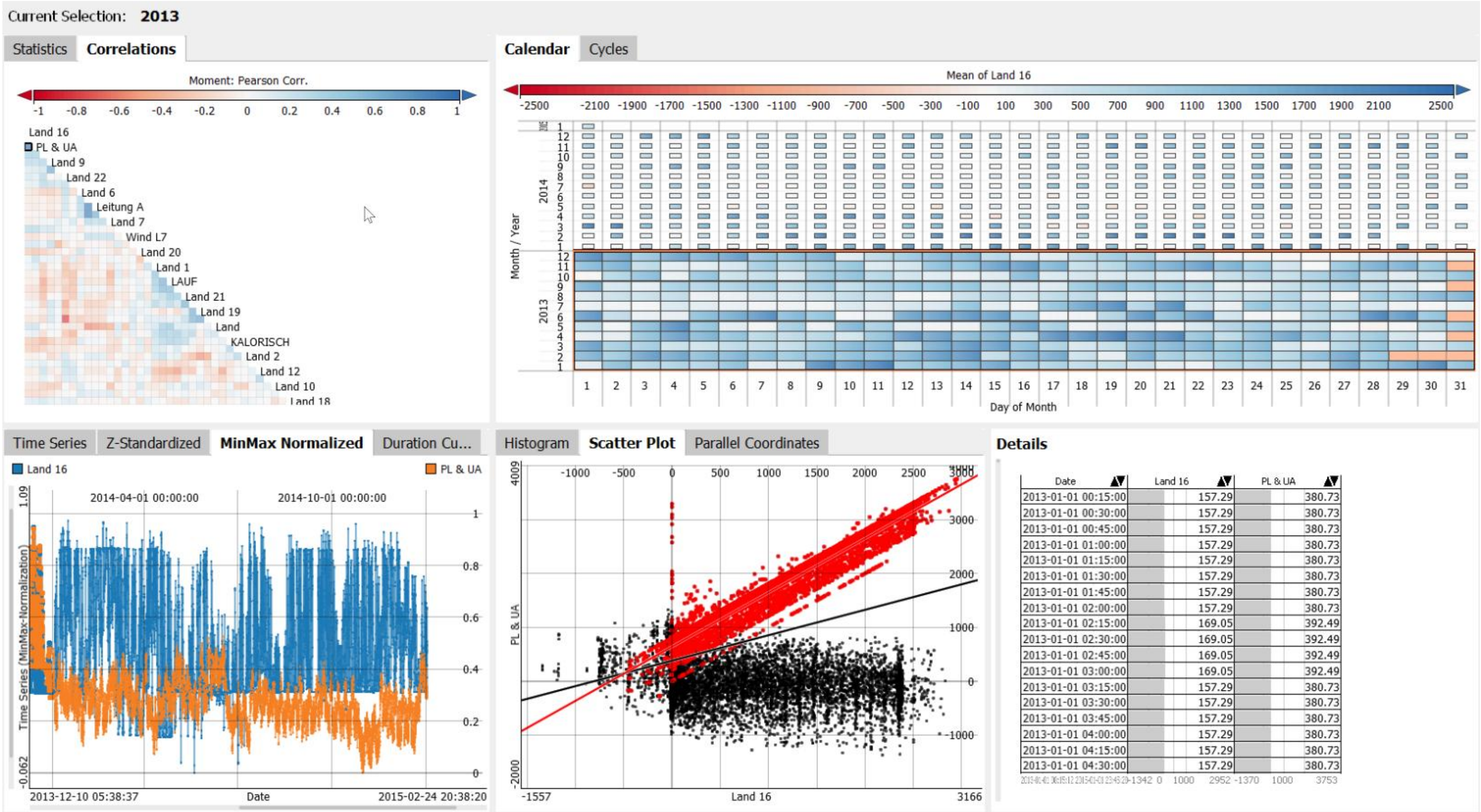
Recurring Questions …

- **Data Quality Assessment and Cleansing**
  Is the data quality sufficient for correct interpretation?
- **Exploratory Data Analysis**
  Does the data reveal unexpected trends and relationships?
- **Model Building and Validation**
  Does the data enable to build (predictive) models?
- **Multi-Criteria Decision Making**
  How to use the data for efficient decision making?
- **Reporting**
  How to effectively communicate insights and decisions?



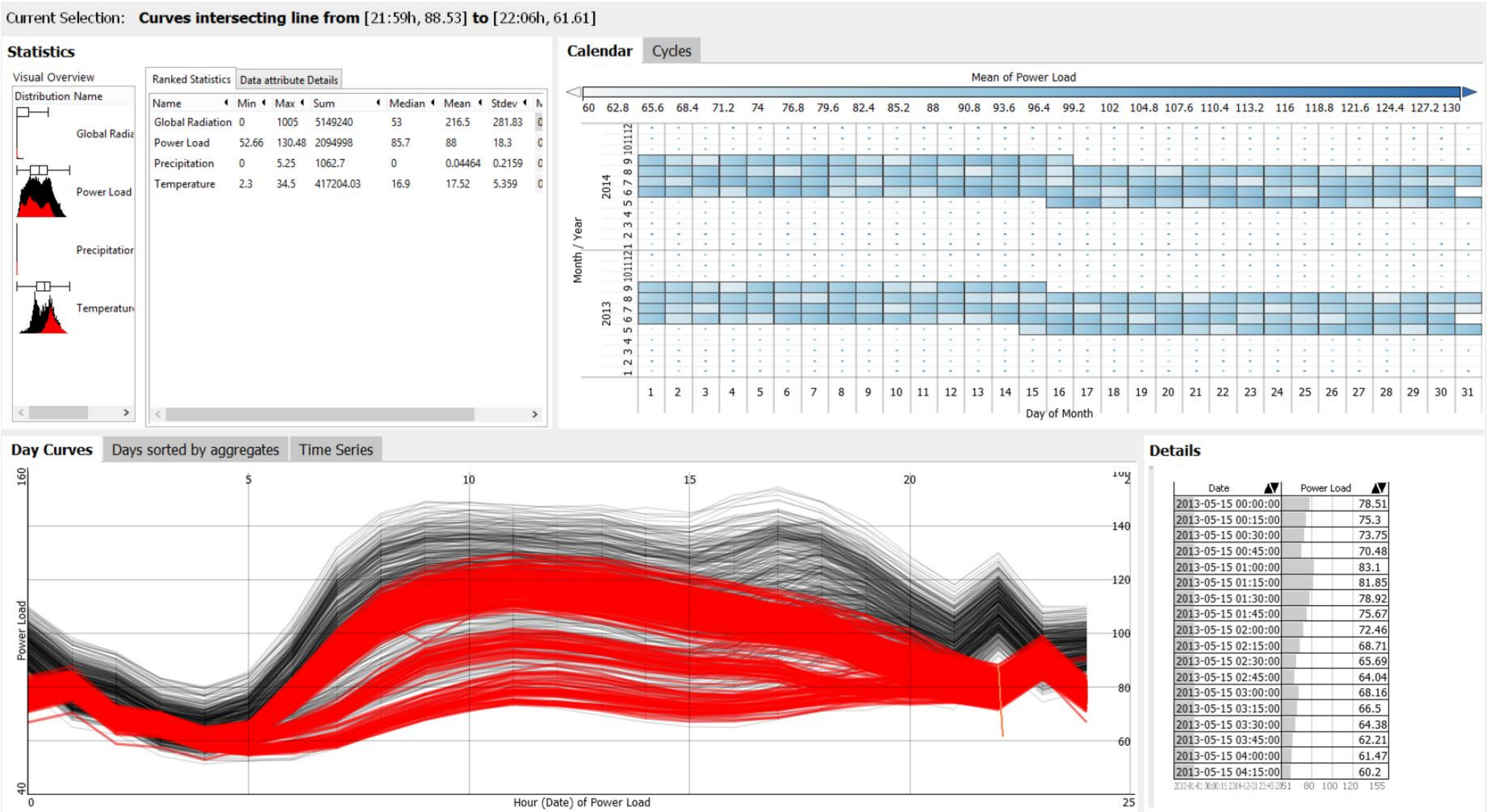
Energy Example:
Exploratory Data Analysis of Times Series Data

- Goal: Detection of structural anomalies and relationships within a few dozens energy time series



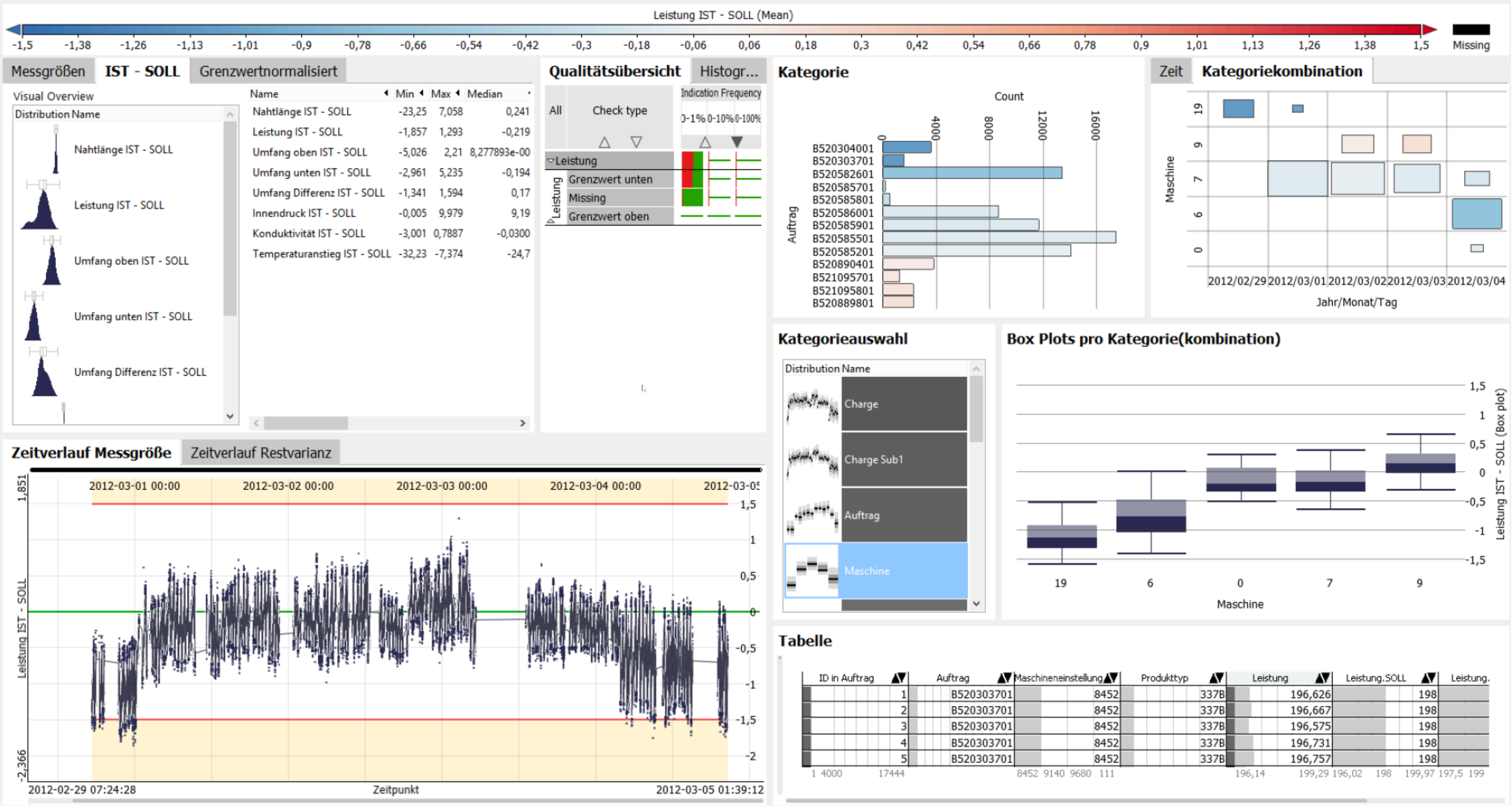
Energy Example:
Detection of Structural Break

- Goal: Explain why automated models show bad performance



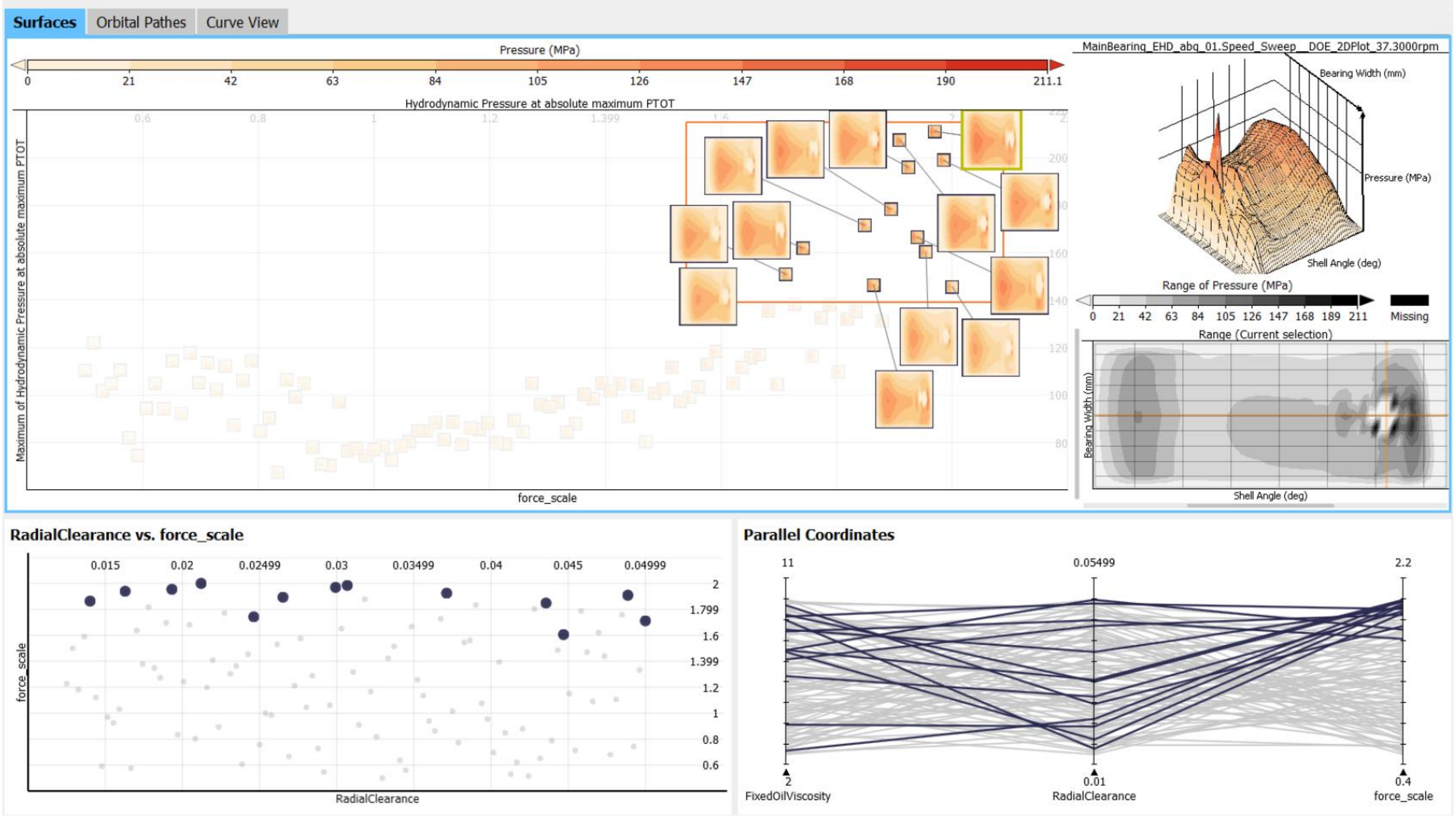
Manufacturing Example: Analysis of Product Quality Indicators

- Goal: Detect and explain quality problems in manufacturing



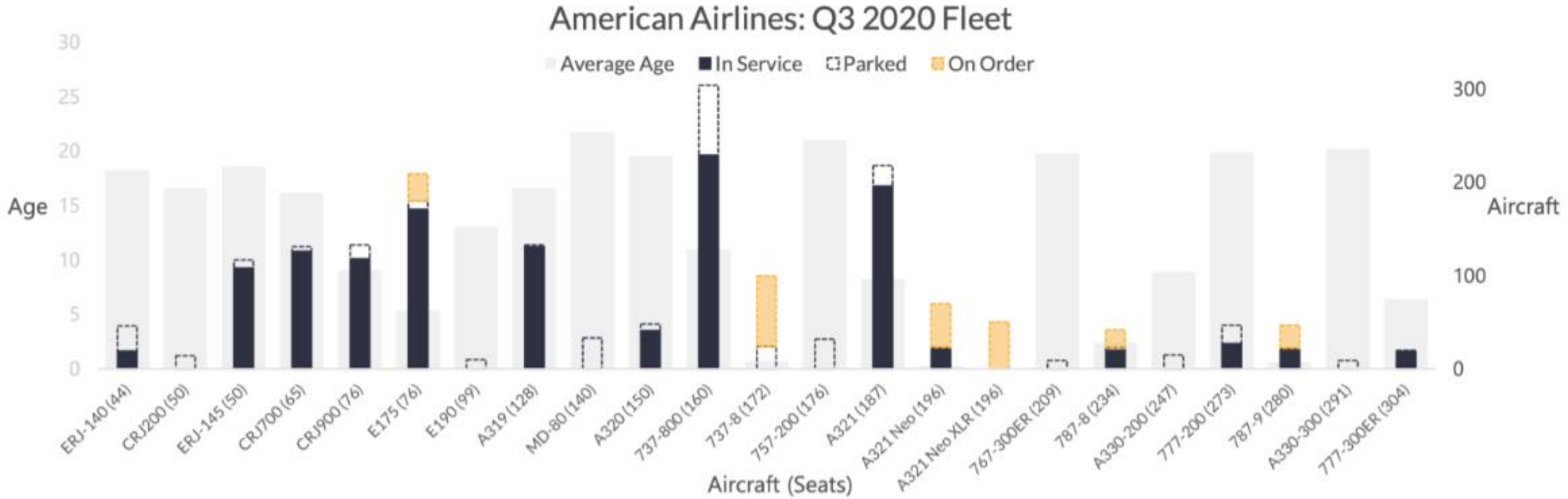
Engineering Example:
Exploratory Analysis of Surface Ensembles

- Goal: Qualitative validation of complex simulation results



Airlines Example: Fleet

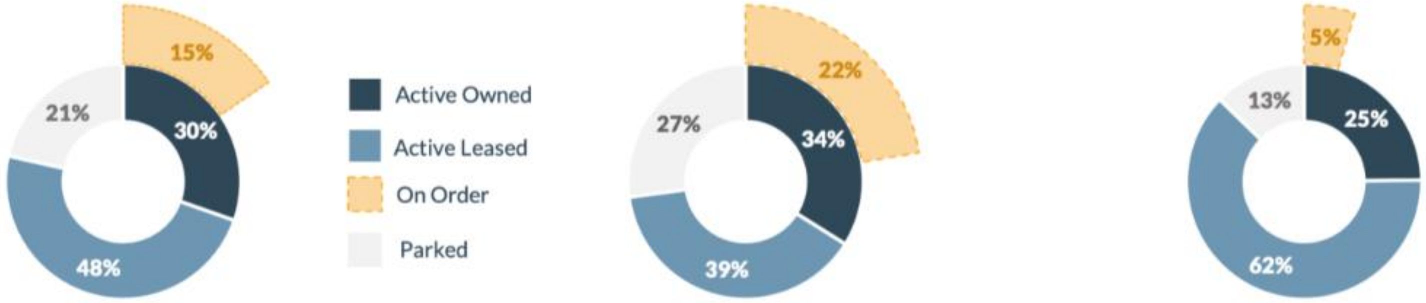
AMERICAN AIRLINES – Fleet



Consolidated	Units	% of Total
Owned - Active	483	30%
Leased - Active	766	48%
Parked	342	21%
On Order	246	15%
In Service	**1249**	**79%**
Total In Service and Storage	**1591**	**-**

Mainline	Units	% of Total
Owned - Active	329	34%
Leased - Active	378	39%
Parked	263	27%
On Order	217	22%
In Service	**707**	**73%**
Total In Service and Storage	**970**	**-**

Regional	Units	% of Total
Owned - Active	154	25%
Leased - Active	388	62%
Parked	79	13%
On Order	29	5%
In Service	**542**	**87%**
Total In Service and Storage	**621**	**-**



Airlines Example: Cash Flow and Operating Statistics

JETBLUE AIRWAYS – Cash Flow and Operating Statistics

CASH FLOW

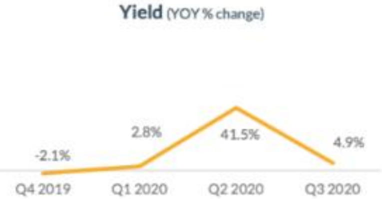
Q3 2020 (Millions)

3 Months Ended September 2020			YOY	8 Quarter Trend
Net Cash Provided by Operating Activities	$	(446)	-293.9%	
Net Cash Provided by Investing Activities	$	(180)	+1900.0%	
Net Cash Provided by Financing Activities	$	60	+328.6%	
Free Cash Flow	$	(892)	-293.9%	
Net Increase in Cash	$	(566)	-340.9%	
Cash at Beginning of Period	$	3,145	+502.5%	
Cash at End of Period	$	2,579	+240.7%	

OPERATING STATISTICS

Q3 2020

Consolidated		YOY	8 Quarter Trend
RPMs (Millions)	2,945	-78.9%	
ASMs (Millions)	6,905	-57.6%	
Load Factor	42.6%	-50.2%	
Yield (Cents)	15.10	+4.9%	
RASM (Cents)	7.12	-44.4%	
CASM (Cents)	14.60	+29.3%	
CASM ex Fuel (Cents)	13.12	+56.3%	
Average Price of Fuel per Gallon (Dollars)	1.23	-40.3%	



Airlines Example: Balance Sheet

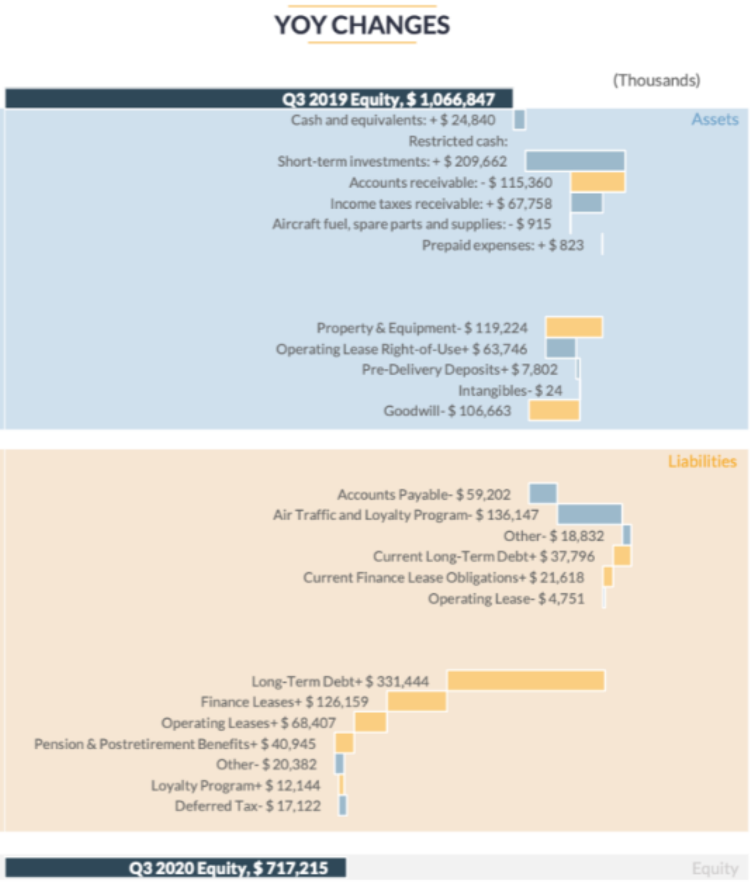
HAWAIIAN AIRLINES – Balance Sheet

Q3 2020 (Thousands)

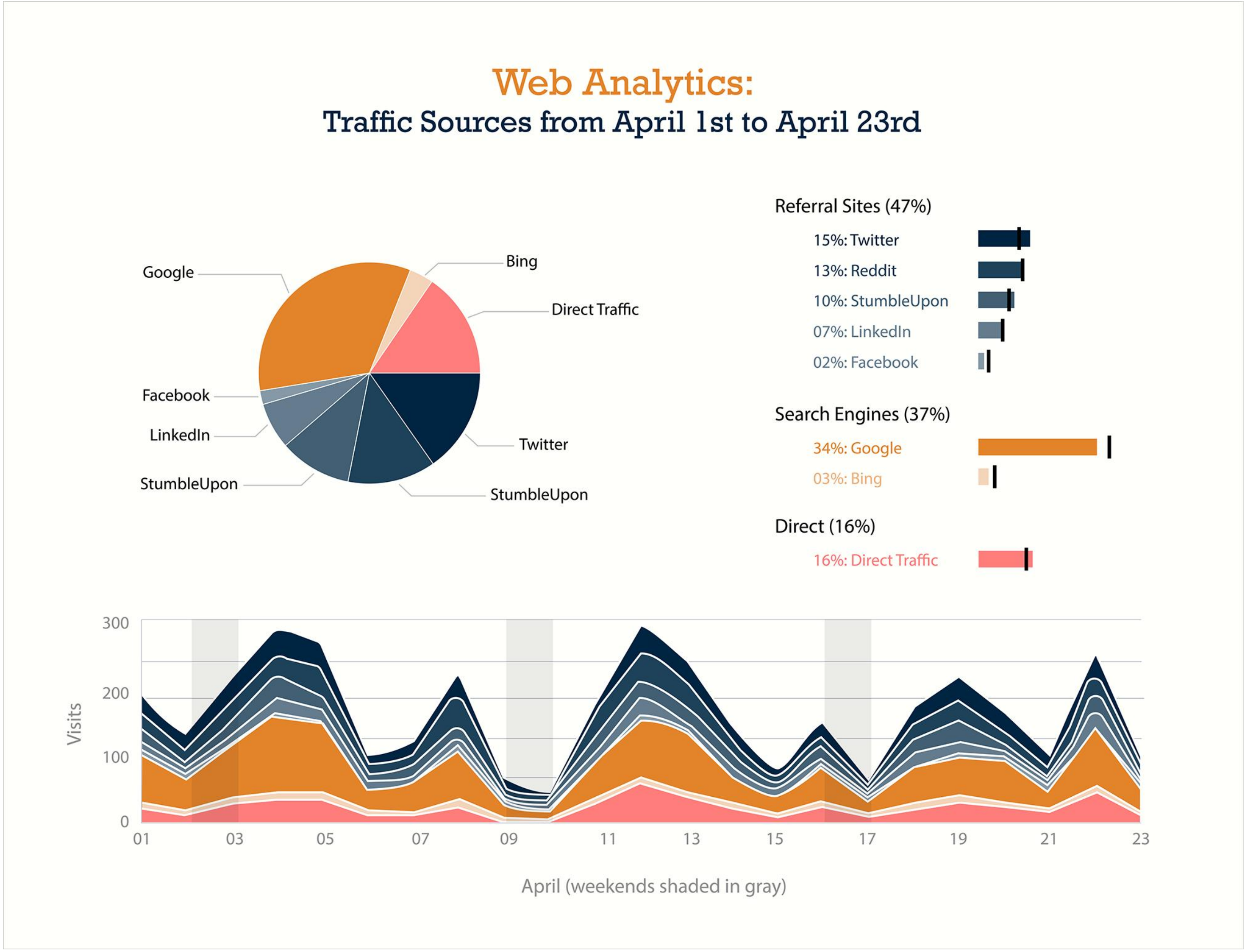
Current Assets		YOY	8 Quarter Trend
Cash and Equivalents	$ 537,002	+4.9%	
Short-Term Investments	$ 442,106	+90.2%	
Accounts Receivable	$ 30,106	-79.3%	
Income Taxes Receivable	$ 67,758	-	
Aircraft Fuel, Spare Parts and Supplies	$ 36,621	-2.4%	
Prepaid Expenses	$ 54,918	+1.5%	
Total Current Assets	**$ 1,168,511**	+19.0%	
Noncurrent Assets			
Property and Equipment	$ 2,121,218	-5.3%	
Operating Lease Right-of-Use	$ 647,288	+10.9%	
Pre-Delivery Deposits on Flight Equipment	$ 146,619	+5.6%	
Intangibles	$ 13,500	-0.2%	
Goodwill	$ -	-100.0%	
Total Noncurrent Assets	**$ 2,928,625**	-5.0%	

Current Liabilities		YOY	8 Quarter Trend
Accounts Payable	$ 96,160	-38.1%	
Air Traffic and Frequent Flyer Deferred Revenue	$ 515,424	-20.9%	
Other Current Liabilities	$ 132,734	-12.4%	
Current Maturities of Long-Term Debt	$ 114,810	+49.1%	
Current Maturities of Finance Lease Obligations	$ 21,618	-	
Operating Lease Current Liabilities	$ 81,881	-5.5%	
Total Current Liabilities	**$ 962,627**	-14.2%	
Noncurrent Liabilities			
Long-Term Debt	$ 1,035,971	+47.0%	
Noncurrent Finance Lease Obligations	$ 126,159	-	
Long-Term Obligations under Operating Leases	$ 524,172	+15.0%	
Accumulated Pensions and Other Post-Retirement Ob	$ 223,907	+22.4%	
Other Noncurrent Liabilities	$ 78,849	-20.5%	
Loyalty Program Deferred Revenue - Noncurrent	$ 186,618	+7.0%	
Deferred Tax Liability	$ 241,618	-6.6%	
Total Noncurrent Liabilities	**$ 2,417,294**	+28.9%	

Shareholders Equity		YOY	8 Quarter Trend
Common Stock	$ 460	-1.5%	
Additional Paid-In Capital	$ 144,884	+8.9%	
Retained Earnings	$ 688,170	-32.8%	
Accumulated Other Comprehensive Loss	$ (116,299)	+29.1%	
Total Equity	**$ 717,215**	-32.8%	



Business Intelligence Example: For Sales and Marketing

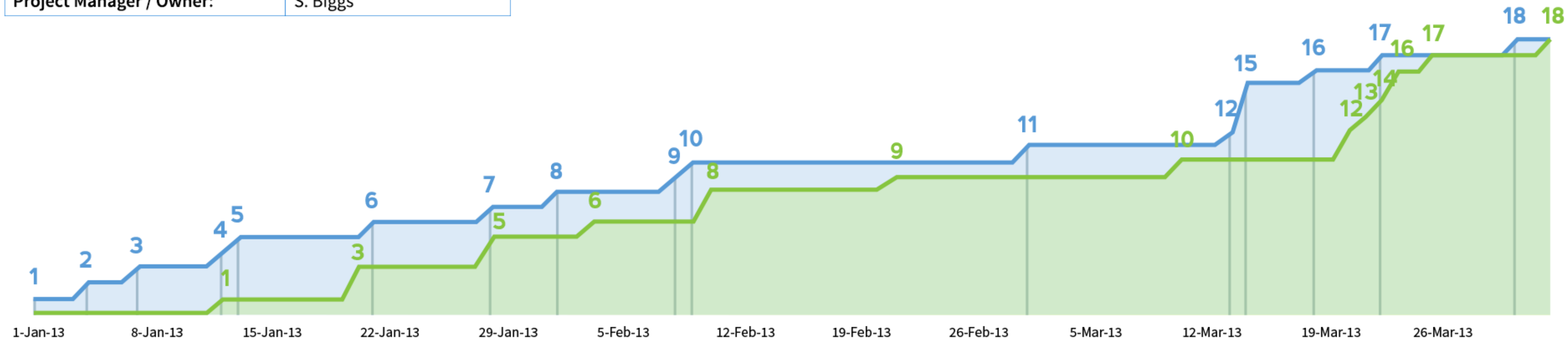


Business Intelligence Example: For Human Resource

Business Intelligence Example: For Tech Team

Project Name:	Project 01
Department:	Office 1234
Project Manager / Owner:	S. Biggs

Open vs. Closed - Issue Tracking & Management

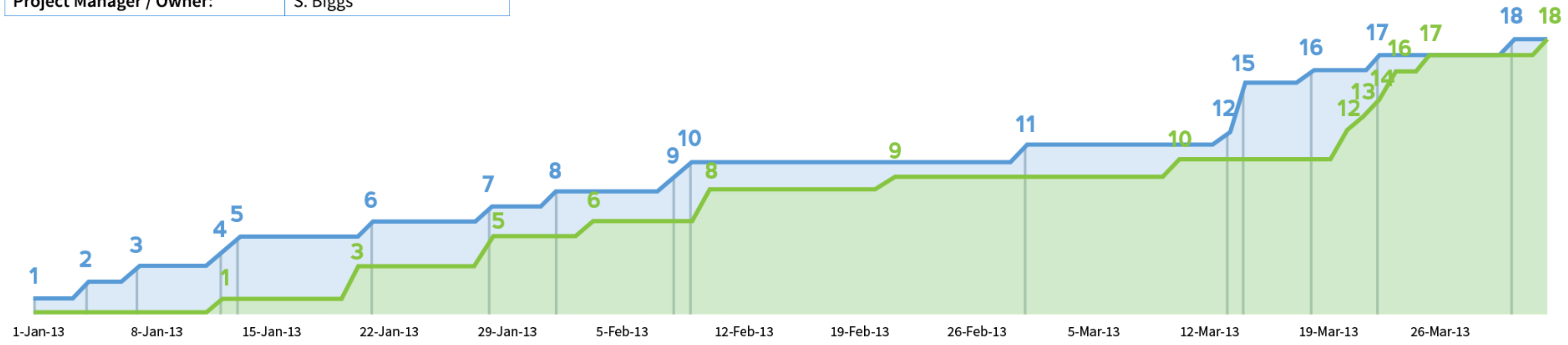


ID	Issue Name	Type	Criticality	Initiator	Status	Created	Closed	Description / Notes
1	Something broke	Type A	High	J. Bob	In Progress	1-Jan-13	20-Jan-13	*This is a note*
2	Something broke	Type B	Medium	R. Brand	In Progress	4-Jan-13	20-Mar-13	*This is another note*
3	Something broke again	Type C	Low	S. Biggs	In Progress	7-Jan-13	21-Feb-13	*This is a really long note that will eventually wrapp around the length of the cell once I get enough content in it.*
4	Something broke	Type A	High	J. Bob	In Progress	12-Jan-13	12-Jan-13	*This is a note*
5	Something broke	Type B	Medium	R. Brand	In Progress	13-Jan-13	20-Jan-13	*This is another note*
6	Something broke	Type C	Low	S. Biggs	In Progress	21-Jan-13	28-Jan-13	*This is another note*
7	Something broke	Type A	High	J. Bob	In Progress	28-Jan-13	28-Jan-13	*This is a note*
8	Something broke	Type B	Medium	R. Brand	In Progress	1-Feb-13	3-Feb-13	*This is another note*
9	Something broke	Type C	Low	S. Biggs	In Progress	8-Feb-13	10-Feb-13	*This is another note*

Business Intelligence Example: For Human Resource

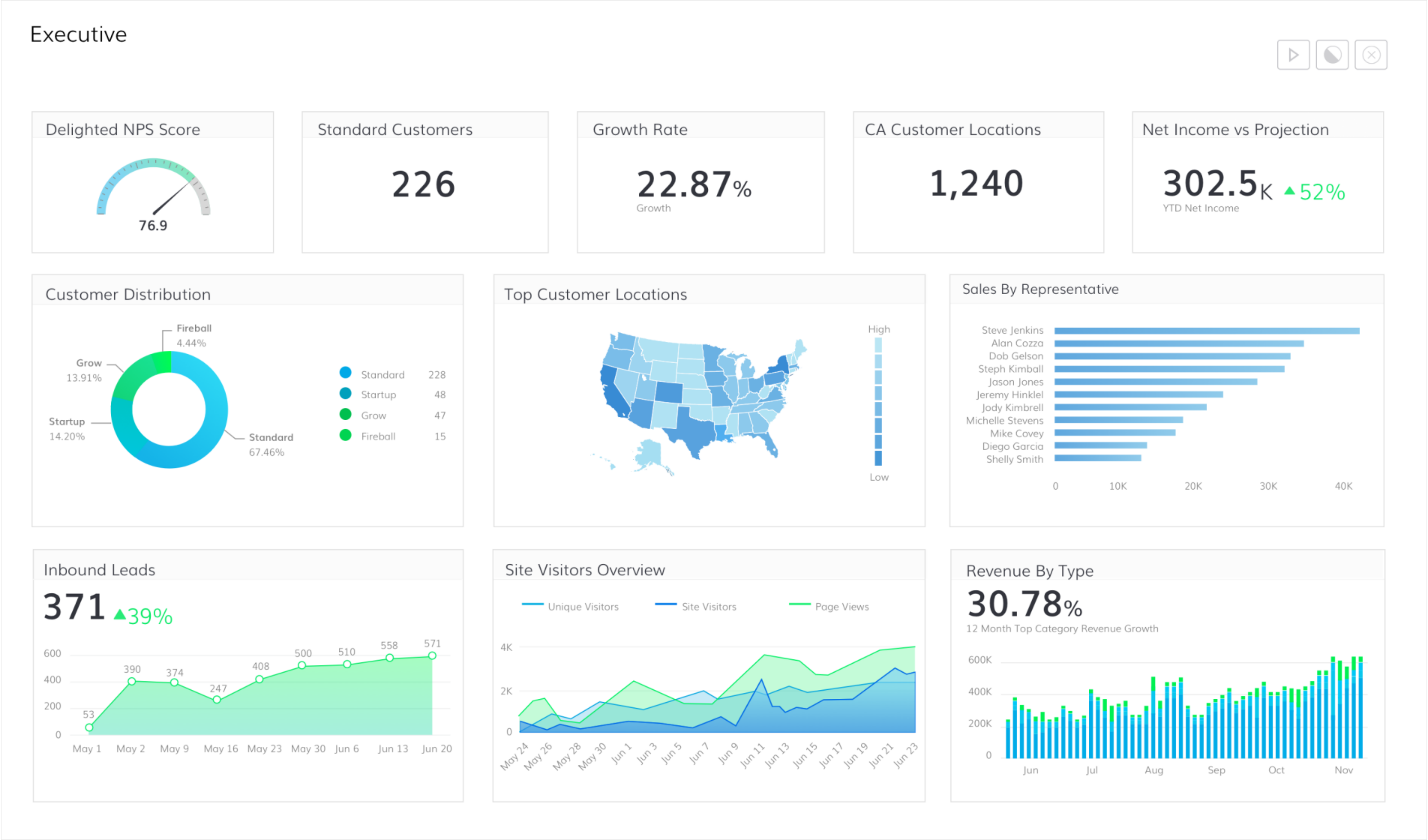
Project Name:	Project 01
Department:	Office 1234
Project Manager / Owner:	S. Biggs

Open vs. Closed - Issue Tracking & Management



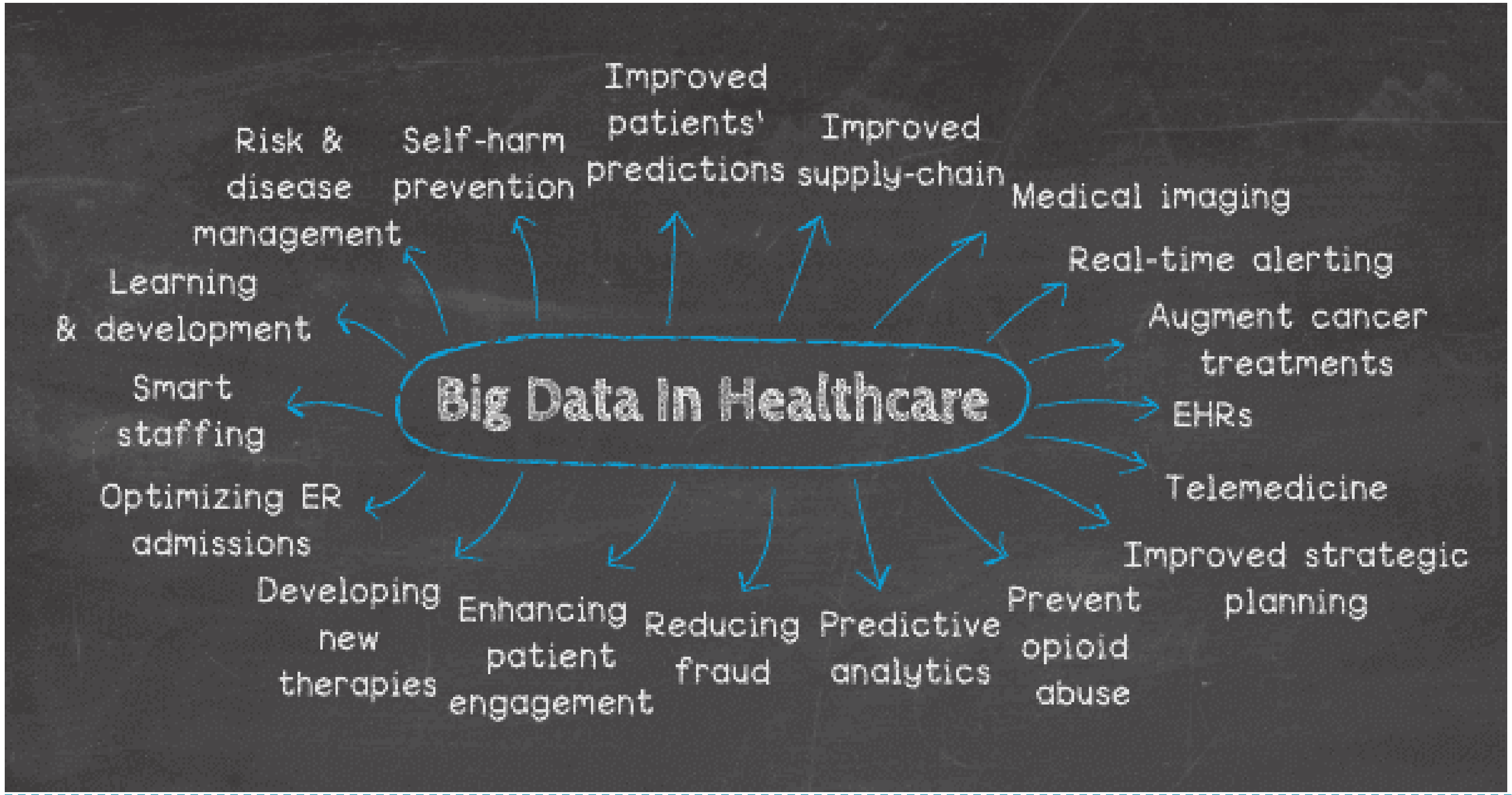
ID	Issue Name	Type	Criticality	Initiator	Status	Created	Closed	Description / Notes
1	Something broke	Type A	High	J. Bob	In Progress	1-Jan-13	20-Jan-13	*This is a note*
2	Something broke	Type B	Medium	R. Brand	In Progress	4-Jan-13	20-Mar-13	*This is another note*
3	Something broke again	Type C	Low	S. Biggs	In Progress	7-Jan-13	21-Feb-13	*This is a really long note that will eventually wrapp around the length of the cell once I get enough content in it.*
4	Something broke	Type A	High	J. Bob	In Progress	12-Jan-13	12-Jan-13	*This is a note*
5	Something broke	Type B	Medium	R. Brand	In Progress	13-Jan-13	20-Jan-13	*This is another note*
6	Something broke	Type C	Low	S. Biggs	In Progress	21-Jan-13	28-Jan-13	*This is another note*
7	Something broke	Type A	High	J. Bob	In Progress	28-Jan-13	28-Jan-13	*This is a note*
8	Something broke	Type B	Medium	R. Brand	In Progress	1-Feb-13	3-Feb-13	*This is another note*
9	Something broke	Type C	Low	S. Biggs	In Progress	8-Feb-13	10-Feb-13	*This is another note*

Business Intelligence Example: For Leadership

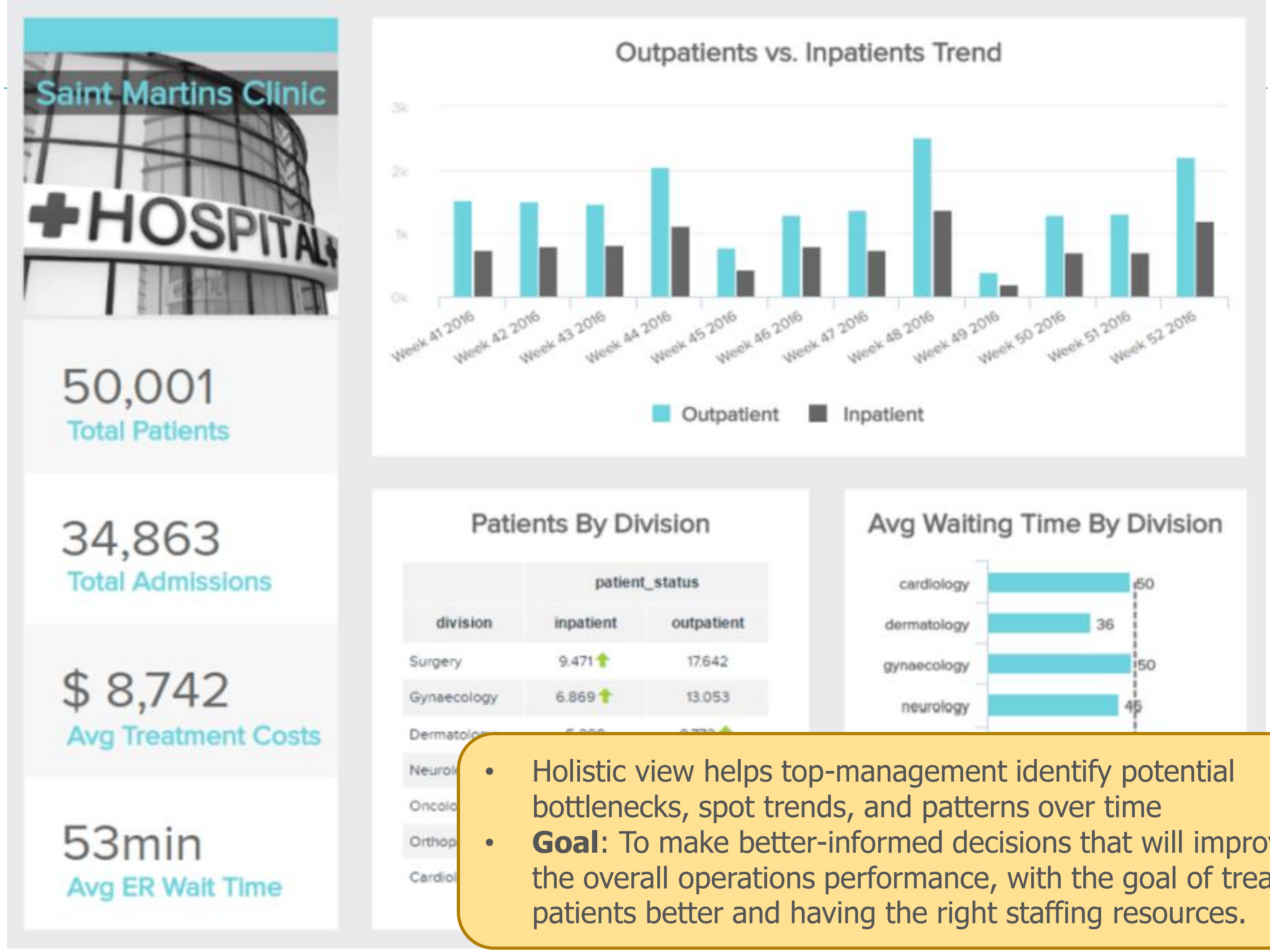


Healthcare Analytics

- Potential to reduce costs of treatment, predict outbreaks of epidemics, avoid preventable diseases, and improve the quality of life in general.



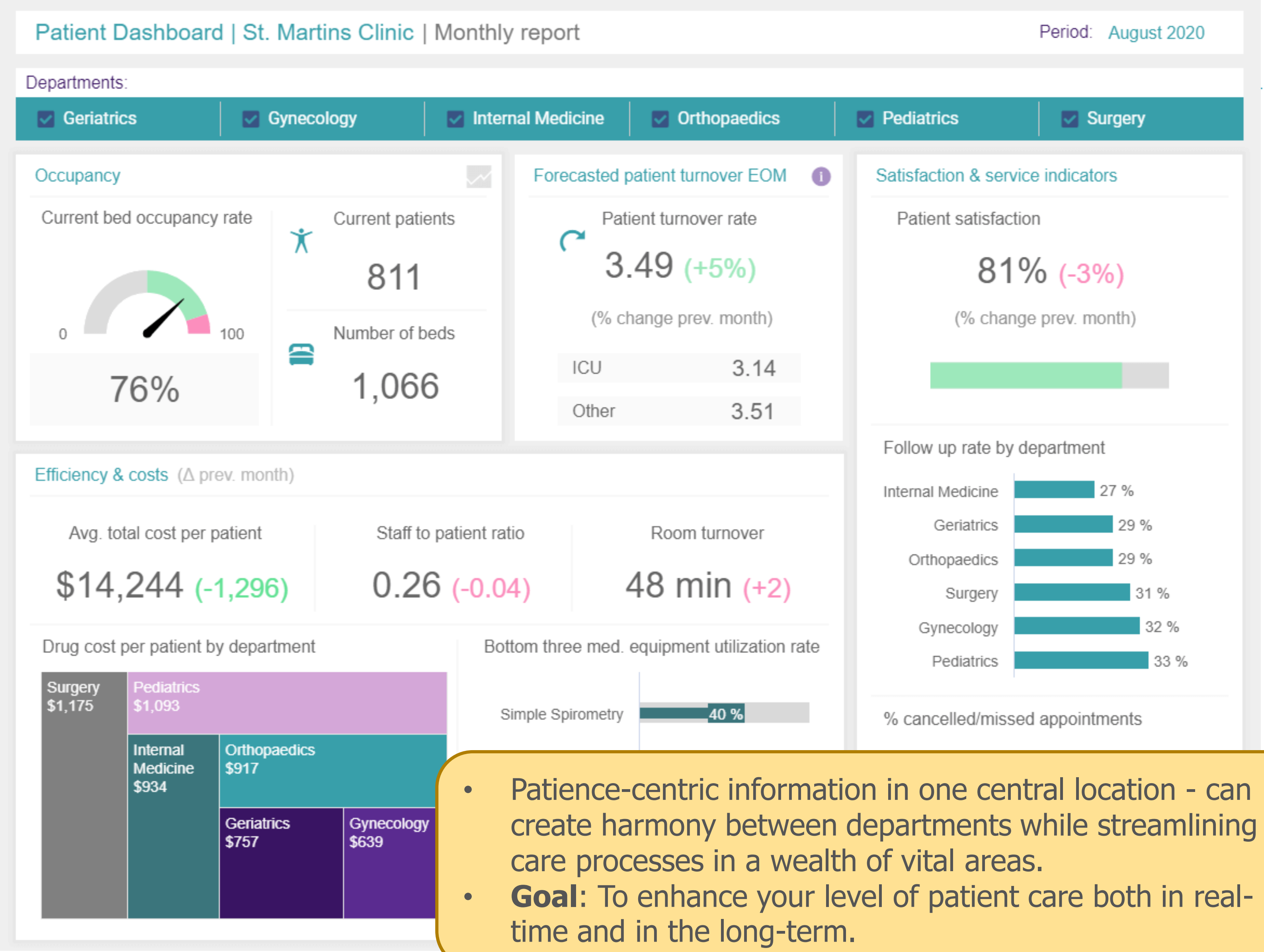
Healthcare Example: Hospital Dashboard



- Holistic view helps top-management identify potential bottlenecks, spot trends, and patterns over time
- **Goal**: To make better-informed decisions that will improve the overall operations performance, with the goal of treating patients better and having the right staffing resources.

https://www.datapine.com/blog/big-data-examples-in-healthcare/

Healthcare Example: Patient Dashboard

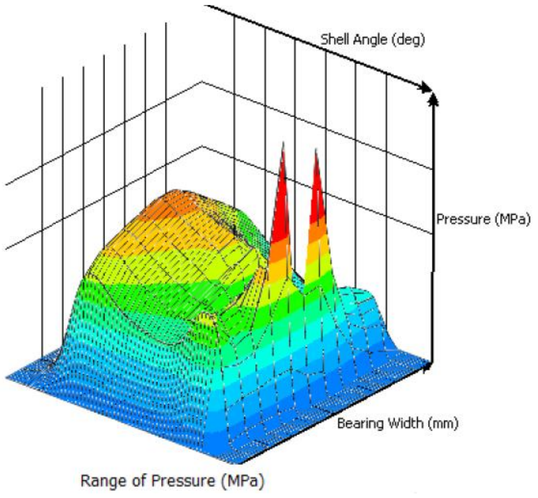


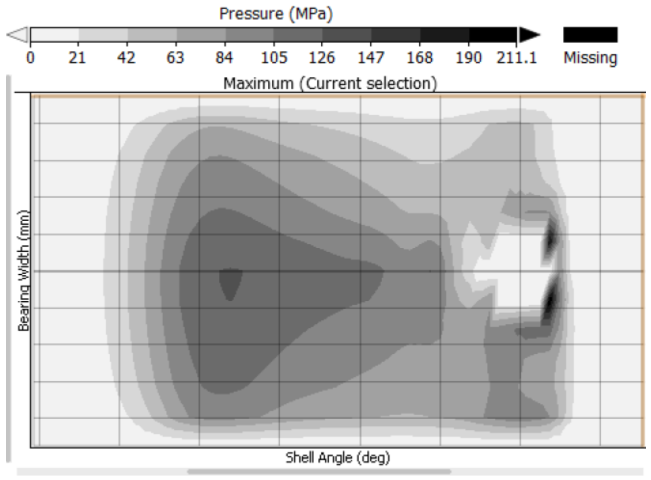
- Patience-centric information in one central location - can create harmony between departments while streamlining care processes in a wealth of vital areas.
- **Goal**: To enhance your level of patient care both in real-time and in the long-term.

https://www.datapine.com/blog/big-data-examples-in-healthcare/

Practical Tips

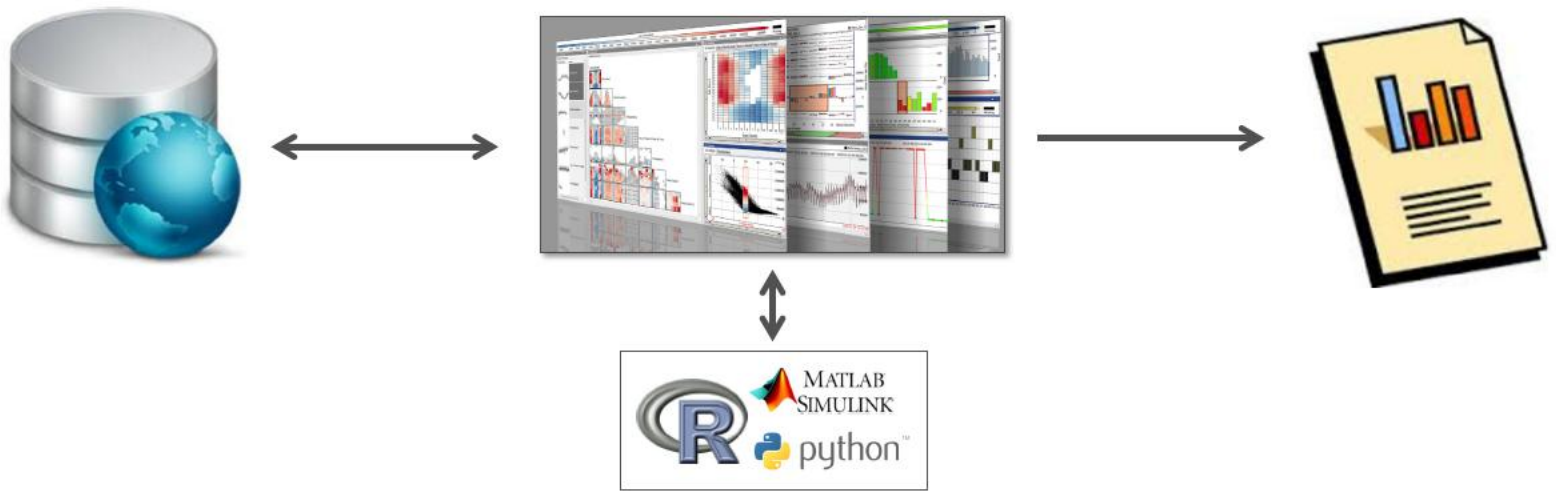
- **The unknown is perceived rather evil than good**
  - Be careful with unfamiliar visualization techniques
  - Be even more careful with complex visualization techniques
  - Real users have very limited time for familiarization





Practical Tips

- **Acceptance requires integration in user work flow**
  - Straightforward data import and result export
  - Real analysts require access to Matlab / Python / R
  - Most users prefer dashboards rather than build-it-yourself kits
  - Automation and guidance is appreciated



Example: Health Care Reform

- https://www.datapine.com/blog/big-data-examples-in-healthcare/
- https://www.webdesignerdepot.com/2009/06/50-great-examples-of-data-visualization/
- https://www.tableau.com/blog/learning-example-real-data-visualizations
- https://www.sas.com/en_my/insights/big-data/data-visualization.html
- https://uncharted.software/assets/business-visualization-applications.pdf
- http://www.ycwu.org/Files/infovis_survey.pdf